Sheet 15 Tue 24 Nov. 2026 *Irrational numbers: $\frac{1}{3}$ is not decimal, $\sqrt{2}$ is irrational*

1. Warm-up · Quick Write $\sqrt{12}$ and $\sqrt{50}$ in the form $a\sqrt{b}$ with b the smallest possible integer.
2. Core · Sorting Among $0.333\ldots$; $\sqrt{9}$; $\sqrt{2}$; π; $\dfrac{22}{7}$, which are rational?
3. Core + · Reasoning Suppose $\sqrt{2} = \dfrac{p}{q}$ with $\dfrac{p}{q}$ in lowest terms, so $p^2 = 2q^2$. What can be said about the parity of p, then of q? What contradiction arises?
4. Review · Table Values 1 ; 2 ; 3 ; 4 with frequencies 5 ; 8 ; 4 ; 3. Give the total frequency, the relative frequency of the value 2 and the mean.

Value	1	2	3	4
Frequency	5	8	4	3

5. Thread · Quick Give the coordinates of the reflection of $A(3;-2)$ in the y-axis.

Sheet 16 Thu 26 Nov. 2026 *Intervals of $\mathbb{R}$: notation and representation*

1. Warm-up · Quick Compare -3 and -5, then $-\dfrac{1}{2}$ and -0.6.
2. Core · Method Write as an interval: $-2 \leqslant x < 5$; $x > 3$; $x \leqslant 0$.
3. Core + · MCQ The intersection $]-\infty;2[\cap [-1;4]$ is:

   A. $]2;4]$ **B.** $]-\infty;4]$ **C.** $[-1;2[$ **D.** $[-1;2]$
4. Review · Table Journey times (min).

Class	$[0;10[$	$[10;20[$	$[20;30[$
Frequency	4	9	7

   What are the total frequency, the modal class and the relative frequency of $[10;20[$?
5. Thread · Problem One bus runs every 15 min, another every 20 min. They are together at 8 a.m. At what time are they next together?

Sheet 17 Mon 30 Nov. 2026 *Absolute value and distance between two real numbers*

1. Warm-up · Quick What is the distance between -3 and 5 on the number line?
2. Core · Quick Calculate $|-7|$, $|3-8|$ and $|\pi - 4|$.
3. Core + · Method Solve $|x-3| = 2$.
4. Review · Formula $f(x) = x^2 - 3$. Give the image of -2 and the preimages of 6.
5. Thread · Formula → table $f(x) = -x + 3$. Complete: $f(-1)$, $f(0)$, $f(2)$.

Sheet 18 Tue 1st Dec. 2026 *Inequality $|x-a| \leqslant r$ and the associated interval*

1. Warm-up · Quick Solve $2x - 3 \leqslant 5$.
2. Core · Method Solve $|x-2| \leqslant 3$ and give S as an interval.
3. Core + · Translate Express as an inequality with absolute value: $x \in [-1;5]$ then $x \in]4;10[$.
4. Review · Problem Pack A: 6 yoghurts for €4.20. Pack B: 10 yoghurts for €6.80. Which is better value?
5. Thread · Quick A disc has radius 5 cm. Give its area and its circumference (exact values).

Functions: representations and reference functions

Sheet 19 Thu 3 Dec. 2026 *Real-valued function; domain of definition*

1. Warm-up · Quick Solve $x - 4 = 0$ then $2x + 6 = 0$.
2. Core · Method Domain of definition of $f(x) = \dfrac{1}{x-4}$?
3. Core + · MCQ The domain of definition of $h(x) = \dfrac{1}{x^2+1}$ is:

   A. $\mathbb{R} \setminus \{1\}$ **B.** $\mathbb{R} \setminus \{-1\,;\,1\}$ **C.** $\mathbb{R}$ **D.** $\mathbb{R}^*$
4. Review · Table Is this table a proportionality table?

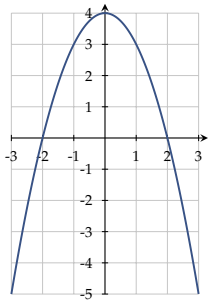
x	2	5	8
y	6	15	24

5. Thread · Code reading What does $f(4)$ return?

```
def f(x):
    return 3*x - 2
```

Sheet 20 Mon 7 Dec. 2026 *Curve with equation $y = f(x)$; images and preimages*

1. Warm-up · Quick $f(x) = -x^2 + 4$. Calculate $f(0)$, $f(-2)$ and $f(3)$.
2. Core · Graph Curve of f:



   Read off $f(1)$ and the preimages of 0.
3. Core + · Thinking For this function, does the equation $f(x) = 5$ have any solutions?
4. Review · Quick Urn: 5 red balls, 3 blue, 2 green. Probability of drawing a blue one? A non-red one?
5. Thread · Method Solve in $\mathbb{R}$: $4 - 3x = 10$.

Sheet 21 Tue 8 Dec. 2026 *Reference functions: absolute value, square, inverse*

1. Warm-up · Quick Calculate $|-2.5|$, $(-3)^2$ and $\dfrac{1}{-4}$.
2. Core · Table Complete, for $x = -2\,;\,-1\,;\,0\,;\,1\,;\,2$, the values of $x \mapsto |x|$, $x \mapsto x^2$ and $x \mapsto \dfrac{1}{x}$.
3. Core + · MCQ Which reference curve passes through $(-1;1)$, $(0;0)$, $(1;1)$ and $(2;4)$?

   A. the line $y = x$ **B.** the inverse function **C.** the absolute value function **D.** the square function
4. Review · Quick An object is enlarged with scale factor 1.5. What does a length of 8 cm become? By what number is the area multiplied?
5. Thread · Prove Conjecture: “the sum of two consecutive integers is odd”. Prove it.